

Hans-Peter Marshall

Associate Professor

Cryosphere Geophysics And Remote Sensing (CryoGARS) group (208) 426-1416 (office)
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Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Professional Experience

- 08/2013-present Associate Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
- 07/2007-present Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
- 08/2008-07/2013 Assistant Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
- 04/2006-08/2008 Research Associate, Inst. of Arctic and Alpine Research (INSTAAR) University of Colorado at Boulder
- 12/2004-04/2006 Research Geophysical Engineer (GS-11), Student Career Experiences Program (SCEP) U.S. Army Corps of Engineers CRREL, Hanover, New Hampshire & Fairbanks, Alaska
- 01/2004-07/2004 Invited visiting scientist, Swiss Federal Institute for Snow and Avalanche Research (SLF), Davos, Switzerland
- 07/2000-12/2005 Research Assistant, INSTAAR, University of Colorado at Boulder
- 11/2002-12/2003 Avalanche Instructor, Alpine World Ascents, Boulder, Colorado
- 06/1995-07/2000 Undergraduate Research Assistant, Glaciology Group, University of Washington
- 08/1999-03/2000 Volunteer high school teacher, Old Crow, Yukon Terr., Canada

Research Funding Summary and Selected External Service

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| 2005-2018 | Awarded/managed ~\$8 mil in research funding from NASA, NSF, DoD, DOTs |
| 2017 | Co-lead for NASA SnowEx, largest seasonal snow remote sensing effort in 15 years |
| 2016 | NASA Earth Venture Mission Panel Member |
| 2015- | Education board member and Incoming Chair, NASA international Snow Working Group for Remote Sensing (iSWGR) |
| 2008-2012 | Cryosphere Section Representative, Fall AGU meeting |
| 2011 | NASA Senior Review Panel Member |

Selected Honors and Awards

2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)
2002-05	N.A.S.A. Earth Systems Science Graduate Fellowship
2003	Arctic Research Consortium of the U.S. (ARCUS) Award for Arctic Research Excellence, Hon. Mention
2001-02	N.S.F. K-12 Engineering Outreach Fellowship
2002	Beverly Sears Graduate Research Grant (University of Colorado)
2001	American Avalanche Association Graduate Research Grant
1999,2001	N.S.F. Graduate Fellowship Program - Honorable Mention
1999	University of Washington Bonderman Honors Travel Fellowship
1999	Volunteer of the Year, Arctic Canada, Frontiers Foundation
1997-99	University of Washington Mary Gates Undergraduate Research Grant
1995-98	N.A.S.A. Space Grant Undergraduate Research Program
1994	Washington Athletic Club Student Athlete of the Year, Nathan Hale High School

Selected Refereed Publications [h-index=19, i10-index=31, citations=1049]

2018	Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., H.P. Marshall , Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. <i>Geophysical Research Letters</i> , 45(7):3164–3172
	Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., H.P. Marshall , Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. <i>IEEE Transactions On Geoscience and Remote Sensing</i> , PP(99):1–15
	Hagenmuller, P., van Herwijnen, A., Pielmeier, C., and H.P. Marshall (2018). Evaluation of the snow penetrometer Avatech SP2. <i>Cold Regions Science and Technology</i> , 149:83–94
2017	Mikesell, T., van Wijk, K., Otheim, L., H.P. Marshall , and Kurbatov, A. (2017). Laser ultrasound observations of mechanical property variations in ice cores. <i>Geosciences</i> , 7(3):47
	Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., H.P. Marshall , and Box, J. (2017). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. <i>The Cryosphere</i> , 11(2):773
	McNamara, J. P., Benner, S. G., Poulos, M. J., Pierce, J. L., Chandler, D. G., Kormos, P. R., H.P. Marshall , Flores, A. N., Seyfried, M., Glenn, N. F., and Aishlin, P. (2017). Form and function relationships revealed by long-term research in a semiarid mountain catchment. <i>WIREs Water</i> , e1267(10.1002/wat2.1267)

- Pegau, W. S., Garron, J., Zabilansky, L., Bassett, C., Bello, J., Bradford, J., Carns, R., Courville, Z., Eicken, H., Elder, B., Eriksen, P., Lavery, A., Light, B., Maksym, T., **H.P. Marshall**, Oggier, M., Perovich, D., Pacwiardowski, P., Singh, H., Tang, D., Wiggins, C., and Wilkinson, J. (2017). Detection of oil in and under ice. *International Oil Spill Conference Proceedings*, 2017(1):1857–1876
- 2016 Evans, S., Flores, A., Heilig, A., Kohn, M., **Marshall, H.P.**, and McNamara, J. (2016). Isotopic evidence for lateral flow and diffusive transport, but not sublimation, in a sloped seasonal snowpack, Idaho, U.S.A. *Geophysical Research Letters*, 43(7):3298–3306
- Rutter, N., **Marshall, H.P.**, Tape, K., Essery, R., and King, J. (2016). Impact of spatial averaging on radar reflectivity at internal snowpack layer boundaries. *Journal of Glaciology*, 62(236):1065–1074
- Bradford, J. H., Babcock, E. L., **Marshall, H.P.**, and Dickins, D. F. (2016). Targeted reflection-waveform inversion of experimental ground-penetrating radar data for quantification of oil spills under sea ice. *Geophysics*, 81(1):WA59–WA70
- 2015 **Marshall, H.P.**, Hawley, R., and Tedesco, M. (2015). *Field measurements for remote sensing of the cryosphere*, chapter 14, pages 345–381. Remote Sensing of the Cryosphere. John Wiley & Sons, Ltd, ed. M. Tedesco
- Arenson, L., Colgan, W., and **Marshall, H.P.** (2015). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands
- Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154
- Kormos, P. R., McNamara, J. P., Seyfried, M. S., **Marshall, H.P.**, Marks, D., and Flores, A. N. (2015). Bedrock infiltration estimates from a catchment water storage-based modeling approach in the rain snow transition zone. *Journal of Hydrology*, 525:231–248
- Hedrick, A., **Marshall, H.P.**, Winstal, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *Cryosphere*, 9(1):13–23
- 2011 **Marshall, H.P.**, Schweizer, J., and Birkeland, K. (2011). Some recent advances in snow and avalanche science. *Cold Regions Science and Technology*, 69(2-3):119–121
- 2009 **Marshall, H.P.** and Johnson, J. B. (2009). Accurate inversion of high-resolution snow penetrometer signals for microstructural and micromechanical properties. *Journal of Geophysical Research-Earth Surface*, 114(F04016)
- 2008 **Marshall, H.P.** and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131
- 2007 **Marshall, H.P.**, Schneebeli, M., and Koh, G. (2007). Snow stratigraphy measurements with high-frequency FMCW radar: Comparison with snow micro-penetrometer. *Cold Regions Science and Technology*, 47(1-2):108–117